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## **Individual Coursework**

### **ST7071CEM: Intelligent Information Retrieval**

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# Introduction

In an era defined by an unprecedented volume of digital information, the ability to efficiently locate relevant data is no longer a convenience but a fundamental necessity. This report provides a comprehensive account of the design, implementation, and rigorous evaluation of a bespoke intelligent information retrieval (IR) system, developed to address this very challenge. The project confronts the complexities of modern data retrieval by strategically bifurcating its focus into two principal components, each tackling a distinct and critical challenge in the field of information science: the creation of a specialized vertical search engine and the development of a high-accuracy document classification system.

Coventry University is developing a vertical search engine for its School of Economics, Finance, and Accounting, focusing on scholarly publications for enhanced precision. It will feature a web crawler, text pre-processing, an inverted index, and TF-IDF ranking.

A document classification system will also categorize unstructured text into "Politics," "Business," or "Health" using a Multinomial Naïve Bayes classifier, automating content filtering and other applications.

This full-stack application prioritizes modularity and scalability. Its backend uses Python with FastAPI (RESTful API), while the frontend is a React and TypeScript SPA with Tailwind CSS and Shadcn/ui for modern styling. The report will detail its architecture, implementation, and evaluation.

# System Architecture

The foundation of this project is a **decoupled client-server architecture**, a paradigm central to modern web development that deliberately separates the presentation layer (frontend) from the business logic and data processing layer (backend). This architectural choice is not merely a design preference but a strategic decision that yields significant, long-term advantages in terms of scalability, maintainability, and development workflow flexibility. By creating a clear boundary between the client and server—with communication occurring exclusively through a well-defined API—each part of the system can be developed, deployed, and scaled independently.

- **Backend (FastAPI Server):** The FastAPI-powered Python backend acts as an API server, handling data requests from authorized clients. It exposes RESTful endpoints like `/search` and `/classify`, performing computationally intensive operations such as web crawling, search indexing, TF-IDF calculations, and machine learning inference. Communication with the frontend is standardized via JSON. Leveraging FastAPI's asynchronous ASGI capabilities, the server efficiently handles concurrent requests, serving many users simultaneously.
- **Frontend (React Client):** The frontend is a sophisticated Single-Page Application (SPA) built with React and TypeScript. Unlike traditional multi-page sites, SPAs load a single HTML shell, dynamically updating content for a fluid, app-like experience without page reloads. This client-side application handles the user interface, rendering components, managing interactions, and client-side state. It communicates with the FastAPI backend via asynchronous API calls for search results or classification predictions, acting purely as a presentation layer. This separation makes the frontend lightweight, focused, and independently maintainable, allowing simultaneous UI/UX and backend development without blocking each other, provided the API contract is consistent.

# Part 1 – Vertical Search Engine

## Crawler

- **Target and Scope:** The crawler's operational domain is strictly and deliberately defined: it is programmed to navigate and extract information solely from the academic publications portal of Coventry University's School of Economics, Finance and Accounting. This focused scope is the defining feature that distinguishes a vertical search engine from a general-purpose one. The crawler is designed to interpret the specific HTML structure of this target site, identifying consistent patterns in the layout to locate lists of publications and the hyperlinks leading to their individual detail pages. This targeted approach ensures that every document in the index is highly relevant to the engine's intended domain.
- **Retrieved Data Fields:** To create a rich, useful, and multifaceted search index, the crawler is configured to extract several key pieces of metadata from each publication's page. The targeted data fields include:
  - **Title:** The full title of the academic paper.
  - **Authors:** A complete list of all contributing authors.
  - **Publication Date:** The year or full date the paper was published, allowing for potential future filtering by date.
  - **Abstract:** The summary of the paper, which provides a dense and highly valuable source of relevant keywords and concepts.
  - **Journal/Conference Name:** The venue where the paper was published.
  - **DOI (Digital Object Identifier):** A persistent and unique link to the publication for easy access and citation.
- **Pre-processing and Data Output:** Raw HTML data is inherently noisy and unsuitable for direct indexing. It contains tags, scripts, and formatting that are irrelevant to the document's content. Therefore, once the data is extracted, it undergoes a crucial pre-processing pipeline to clean, normalize, and structure it. This pipeline consists of several standard Natural Language Processing (NLP) steps, each serving a specific purpose:

- **Lowercasing:** All text is converted to lowercase to ensure case-insensitivity. This prevents the system from treating terms like "Market" and "market" as two different words.
- **Tokenization:** The continuous stream of text is broken down into individual words or "tokens." This is the fundamental step in preparing text for any kind of computational analysis.
- **Stop-Word Removal:** Common words that carry little semantic meaning (e.g., "the," "a," "is," "in," "of") are identified and removed from the token list. This reduces the size of the index and improves relevance by focusing on more meaningful terms.
- **Stemming:** Tokens are reduced to their morphological root form (e.g., "investing," "invested," and "investment" all become "invest"). This helps group related words and improves recall, ensuring that a search for "invest" can retrieve documents containing any of its variations. The cleaned, processed data for all publications is then structured into a standardized format, typically a list of JSON objects, where each object represents a single document, ready for the indexing phase.
- **Crawler Operation:** The crawler is implemented as a Python script (crawler.py) using the requests library for making HTTP requests to fetch web pages and BeautifulSoup for parsing the resulting HTML. A core principle of its design is **"politeness."** It programmatically respects the robots.txt file of the target server, which outlines rules for automated agents. Furthermore, it employs rate-limiting by introducing a small, deliberate delay (e.g., using time.sleep()) between consecutive requests to avoid overwhelming the university's server with rapid-fire traffic. To ensure the search index remains current with new publications, the crawling process is automated. It is designed to be run as a **weekly scheduled task**. This can be managed at the operating system level using a utility like cron on Linux/macOS or within the application itself using a Python library like APScheduler, which can provide more fine-grained control over scheduling. The script includes robust error handling to gracefully manage issues like broken links or temporary server issues (5xx errors), ensuring the crawling process is resilient

and can run unattended.



```

# Step 2: Visit each publication link and scrape details
for i, pub_info in enumerate(publication_details_list):
    link = pub_info['link']
    title = pub_info['title']
    print(f"Scraping details for publication {i +
1}/{len(publication_details_list)}: {link}")

    title_final, authors, date, abstract =
scrape_publication_details(driver, link, title)

    publications_data.append({
        'id': i, # Assigning a simple ID
        'title': title_final,
        'authors': json.dumps(authors), # Serialize authors to a
string for CSV
        'date': date,
        'abstract': abstract,
        'publication_link': link
    })

    time.sleep(POLITE_DELAY)

return publications_data

except TimeoutException:
    print("Page load timed out. Some content might not have been
retrieved.")
    return publications_data
except WebDriverException as e:
    print(f"WebDriver error during crawl: {e}")
    return publications_data
finally:
    if driver:
        driver.quit()

```



```

def scrape_publication_details(driver, url, title_from_list):
    """Scrapes authors and abstract from a single publication page."""
    try:
        driver.get(url)
        WebDriverWait(driver, 10).until(
            EC.presence_of_element_located((By.CSS_SELECTOR, '.container')))

        soup = BeautifulSoup(driver.page_source, 'html.parser')

        # Use the title scraped from the list page
        title = title_from_list
        authors, date, abstract = [], 'N/A', ''

        # Scrape authors
        try:
            authors = fetch_authors(soup, BASE_URL)
        except Exception as e:
            print(f"Could not find authors on {url}: {e}")

        # Scrape abstract
        abstract = fetch_abstract(soup)

        # Scrape date
        date_tag = soup.find('span', class_='date')
        date = date_tag.text.strip() if date_tag else 'N/A'

        return title, authors, date, abstract

    except (TimeoutException, WebDriverException) as e:
        print(f"Error loading page {url}: {e}")
        return title_from_list, [], 'N/A', ''

```



```

def fetch_authors(soup, base_url):
    authors_data = []
    persons_p = soup.select_one('p.relations.persons')
    if not persons_p:
        return []
    for element in persons_p.contents:
        if isinstance(element, Tag) and element.name == 'a':
            name = element.get_text(strip=True)
            url = urljoin(base_url, str(element.get('href', '')))
            if name:
                authors_data.append({'name': name, 'url': url})
        elif isinstance(element, str):
            potential_names = element.split(',')
            for name_part in potential_names:
                clean_name = name_part.strip(' ,')
                if clean_name:
                    authors_data.append({'name': clean_name, 'url': None})
    return authors_data

```



```

def fetch_abstract(soup):
    # Find the specific div containing the abstract
    abstract_div = soup.find('div',
class_='rendering_researchoutput_abstractportal')
    if abstract_div:
        # The text is within a nested 'textblock' div
        text_block = abstract_div.find('div', class_='textblock')
        if text_block:
            return text_block.get_text(strip=True)
    return '' # Return empty string if abstract is not found

```

# Indexer

- **Implementation Method:** The indexer (indexer.py) is the component responsible for building the data structure that enables fast and efficient search. It is designed to be triggered automatically upon the successful completion of a crawl, creating a seamless data pipeline that takes information from the web and integrates it into the search engine's core.
- **Data Structures:** The **inverted index** is the core of search engines, enabling efficient retrieval. Like a textbook index, it lists unique words (terms) and their associated **posting lists**, which detail the documents where each term appears, often including document ID and term frequency. This structure allows quick identification of documents containing specific query terms.term with a single lookup, rather than having to scan every document in the corpus.
- **TF-IDF Model and Index Construction:** The indexer builds the inverted index and pre-calculates TF-IDF scores for ranking. TF-IDF is a statistical measure of a word's importance to a document in a corpus. Its importance grows with the word's frequency in the document but is offset by its frequency in the corpus. This has two parts:
  - **Term Frequency (TF):** This measures how frequently a term  $t$  appears in a document  $d$ . A higher TF suggests the term is important to that specific document. To prevent a bias towards longer documents, this is often normalized. The formula is:
$$TF(t,d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$
  - **Inverse Document Frequency (IDF):** This measures how important a term is across the entire collection of documents  $D$ . The IDF of a rare word is high, whereas the IDF of a common word (like "data" or "analysis" in a technical corpus) is low. The logarithm is used to dampen the effect of very high IDF values, preventing rare words from completely dominating the score. The formula is:
$$IDF(t,D) = \log\left(\frac{\text{Total number of documents } D}{\text{Number of documents containing term } t}\right)$$
  - The final **TF-IDF score** for a term  $t$  in a document  $d$  from corpus  $D$  is the

product of these two values:  $TF\text{-}IDF(t,d,D) = TF(t,d) \times IDF(t,D)$

These scores form the basis of the vector space model, where each document is represented as a vector of its TF-IDF scores, enabling quantitative comparisons of relevance.

```
def build_index():
    """
    Builds an enhanced inverted index from a CSV file of documents.
    """
    try:
        document_store = []
        with open(PUBLICATIONS_FILE, 'r', encoding='utf-8') as f:
            reader = csv.DictReader(f)
            for row in reader:
                document_store.append(row)
    except FileNotFoundError:
        print(f"Error: {PUBLICATIONS_FILE} not found. Ensure the CSV file exists.")
        return

    if not document_store:
        print("Warning: The publications file is empty.")
        return

    print(f"Building enhanced inverted index from {len(document_store)} documents...")

    inverted_index = defaultdict(dict)
    document_metadata = {}
```

```

FIELD_WEIGHTS = {'title': 5.0, 'keywords': 3.0, 'authors': 2.5, 'abstract': 1.0}
current_year = datetime.now().year

for doc_id, document in enumerate(document_store):
    doc_id_str = str(doc_id)
    title = document.get('title', '').strip()
    if not title:
        continue

    # Process text from fields
    title_tokens = preprocess_text(title)
    abstract = document.get('abstract', '')
    abstract_tokens = preprocess_text(abstract)

    # Safely parse authors from string to list of names
    try:
        authors_data = ast.literal_eval(document.get('authors', '[]'))
        if authors_data and isinstance(authors_data[0], dict):
            authors = [author.get('name', '') for author in authors_data]
        else:
            authors = authors_data
    except (ValueError, SyntaxError):
        authors = []
    author_tokens = preprocess_text(" ".join(authors))

    # Safely parse keywords
    try:
        keywords_data = ast.literal_eval(document.get('keywords', '[]'))
        keywords = keywords_data if isinstance(keywords_data, list) else []
    except (ValueError, SyntaxError):
        keywords = []
    keyword_tokens = preprocess_text(" ".join(keywords))

    # Calculate weighted term frequencies
    term_frequencies = defaultdict(float)
    for token in title_tokens:
        term_frequencies[token] += FIELD_WEIGHTS['title']
    for token in keyword_tokens:
        term_frequencies[token] += FIELD_WEIGHTS['keywords']
    for token in author_tokens:
        term_frequencies[token] += FIELD_WEIGHTS['authors']
    for token in abstract_tokens:
        term_frequencies[token] += FIELD_WEIGHTS['abstract']

```

```

pub_year = None
date_str = document.get('date', '').strip()

# MODIFICATION: Standardize "Sept" to "Sep" before parsing
date_str = date_str.replace('Sept', 'Sep')

if date_str:
    possible_formats = ['%d %b %Y', '%b %Y', '%Y']
    for fmt in possible_formats:
        try:
            pub_year = datetime.strptime(date_str, fmt).year
            break
        except ValueError:
            continue
    if not pub_year:
        print(f"Warning: Could not parse date '{date_str}' for doc_id {doc_id}
with any known format.")

if pub_year and pub_year > 1900:
    years_since = current_year - pub_year
    if 0 <= years_since <= 5:
        boost = 1.0 + (0.2 * (5 - years_since) / 5)
        for token in term_frequencies:
            term_frequencies[token] *= boost

# Normalize term frequencies
doc_length = sum(term_frequencies.values())
if doc_length > 0:
    for token, tf in term_frequencies.items():
        inverted_index[token][doc_id_str] = tf / doc_length

# Store metadata for filtering and display
document_metadata[doc_id_str] = {
    'title': title,
    'url': document.get('publication_link', ''),
    'year': pub_year,
    'authors': authors
}

```

```
# Calculate IDF for each term
num_docs = len(document_metadata)
final_index = {}
for term, postings in inverted_index.items():
    doc_freq = len(postings)
    idf = math.log(1 + num_docs / (1 + doc_freq))
    final_index[term] = {'idf': idf, 'postings': postings}

# Save the complete index to a file
index_data = {
    'index': final_index,
    'metadata': document_metadata,
    'stats': {'total_docs': num_docs, 'total_terms': len(final_index)}
}
with open(INDEX_FILE, 'w', encoding='utf-8') as f:
    json.dump(index_data, f, indent=2, ensure_ascii=False)

print("✅ Indexing complete!")
print(f" 📊 {len(final_index)} unique terms from {num_docs} documents.")
print(f" 💾 Index saved to {INDEX_FILE}")
```

# Query Processor

- **Supported Query Types:** The system is designed to handle **free-text queries**. This means users can enter a natural string of keywords (e.g., "machine learning models for financial forecasting"), and the system will interpret this query to return the most relevant documents. Multi-word queries are processed using the vector space model, where the query itself is treated as a mini-document and transformed into its own TF-IDF vector.
- **Ranked Retrieval Method:** The **Vector Space Model (VSM)** is central to ranked retrieval systems. It represents documents and queries as vectors, where each dimension corresponds to a unique term and its TF-IDF score. Document relevance is determined by **cosine similarity** between query and document vectors, with higher scores indicating greater similarity (0 to 1). Documents are ranked by this score, most relevant first.
- **Process:** The process of handling a user query is a well-defined, multi-step operation:
  1. **Query Reception:** The FastAPI backend receives a query string from the React frontend via the /search endpoint.
  2. **Query Pre-processing:** The raw query string undergoes the exact same pre-processing pipeline as the documents (lowercasing, tokenization, stop-word removal, stemming). This is a critical step to ensure that the terms in the query can be matched against the terms in the index.
  3. **Vectorization:** The processed query is converted into a TF-IDF vector using the pre-calculated IDF values from the indexing phase.
  4. **Candidate Document Retrieval:** The query processor uses the inverted index to quickly identify a set of candidate documents—only those documents that contain at least one of the terms from the query need to be considered for ranking. This is a massive efficiency gain over a linear scan.
  5. **Similarity Calculation:** For each candidate document, the cosine similarity between its TF-IDF vector and the query's TF-IDF vector is calculated.
  6. **Ranking:** The documents are sorted based on their cosine similarity scores, from highest to lowest.



7. **Response Generation:** The top-ranked documents (e.g., the top 10) are formatted into a JSON response, including all relevant metadata (title, authors, abstract, etc.), and sent back to the frontend for display.

```
class SearchRequest(BaseModel):
    query: str

class Publication(BaseModel):
    """Represents a single publication document in the response."""
    model_config = ConfigDict(from_attributes=True)

    title: Optional[str] = None
    authors: List[str] = []
    date: Optional[str] = None
    abstract: Optional[str] = None
    publication: Optional[str] = None
    publicationYear: Optional[int] = None
    category: Optional[str] = None
    publicationUrl: Optional[str] = Field(None, alias='publication_link')
    relevanceScore: Optional[float] = None

class SearchResponse(BaseModel):
    total: int
    publications: List[Publication]
```

```

def _calculate_tf_idf_scores(query_tokens: List[str]) -> DefaultDict[str, float]:
    """Calculates TF-IDF scores for documents based on query tokens."""
    scores = defaultdict(float)
    for token in query_tokens:
        if token in index_data:
            term_info = index_data[token]
            idf = term_info.get('idf', 1.0)
            for doc_id, tf in term_info['postings'].items():
                scores[doc_id] += tf * idf
    return scores

```

```

def _filter_doc_ids_by_year(
    doc_ids: List[str],
    min_year: Optional[int],
    max_year: Optional[int]
) -> List[str]:
    """Filters a list of document IDs based on a year range."""
    if not min_year and not max_year:
        return doc_ids

    filtered_ids = []
    for doc_id in doc_ids:
        meta = document_metadata.get(doc_id)
        if not meta or meta.get('year') is None:
            continue

        pub_year = meta['year']
        if (min_year and pub_year < min_year) or (max_year and pub_year >
max_year):
            continue

        filtered_ids.append(doc_id)
    return filtered_ids

```

```

def _get_classified_publications(
    doc_ids: List[str],
    scores: DefaultDict[str, float]
) -> List[dict]:
    """Retrieves and classifies full publication data for a list of IDs."""
    results = []
    for doc_id in doc_ids:
        try:
            # Create a copy to avoid modifying the global document_store
            doc = document_store[int(doc_id)].copy()
            doc['relevanceScore'] = scores.get(doc_id, 0.0)
            text_to_classify = (doc.get('title', '') + ' ' + doc.get('abstract',
            '')).strip()

            if classifier and vectorizer and text_to_classify:
                processed_text = preprocess_for_classification(text_to_classify)
                vectorized_text = vectorizer.transform([processed_text])
                predicted_category = classifier.predict(vectorized_text)[0]
                doc['category'] = predicted_category
            else:
                doc['category'] = 'Unclassified'

            results.append(doc)
        except (IndexError, ValueError) as e:
            print(f"Warning: Could not retrieve document for ID '{doc_id}'. Error: {e}")
            continue
    return results

```

```

@app.post("/api/search", response_model=SearchResponse)
async def search_publications(
    request: SearchRequest,
    # MODIFIED: Removed page and page_size parameters
    min_year: Optional[int] = None,
    max_year: Optional[int] = None
):
    """
    Searches publications based on a query, filters by year,
    and returns all matching, classified results.
    """
    print("-" * 50)
    print("🚀 NEW SEARCH REQUEST RECEIVED 🚀")

    print("\n[INPUT] Request Body (SearchRequest):", request)
    print(f"[INPUT] Query Parameters: min_year={min_year}, max_year={max_year}")

    if not index_data or not document_store:
        raise HTTPException(status_code=503, detail="Search index is not available.")

    # 1. Process query and calculate scores
    query_tokens = preprocess_for_search(request.query)
    print("\n[STEP 1] Preprocessed Query Tokens:", query_tokens)
    scores = _calculate_tf_idf_scores(query_tokens)
    print("\n[STEP 1] Calculated TF-IDF Scores (first 5):")
    pprint.pprint(dict(list(scores.items())[:5]))

    if not scores:
        print("\n[INFO] No documents matched the query. Returning empty response.")
        return SearchResponse(total=0, publications=[])

    # 2. Filter results by year
    doc_ids = list(scores.keys())
    print(f"\n[STEP 2] Document IDs before year filtering: {len(doc_ids)} total")
    filtered_doc_ids = _filter_doc_ids_by_year(doc_ids, min_year, max_year)
    print(f"[STEP 2] Document IDs AFTER year filtering: {len(filtered_doc_ids)} total")

    # 3. Sort by relevance score
    sorted_doc_ids = sorted(
        filtered_doc_ids,
        key=lambda doc_id: scores[doc_id],
        reverse=True
    )
    print("\n[STEP 3] Sorted Document IDs (first 10):", sorted_doc_ids[:10])

    # 4. MODIFIED: Pagination is removed. All results will be returned.
    total_results = len(sorted_doc_ids)
    print(f"\n[STEP 4] Found {total_results} total matching results. Returning all.")

    # 5. Retrieve full publication data for ALL sorted IDs and classify
    results = _get_classified_publications(sorted_doc_ids, scores)
    print("\n[STEP 5] Final classified publications being returned (first result):")
    if results:
        pprint.pprint(results[0])
    else:
        print("[STEP 5] No results to return.")

    # Convert dicts to Publication objects
    publications = [Publication(**doc) for doc in results]

    print("\n✅ REQUEST COMPLETE")
    print("-" * 50)

    return SearchResponse(total=total_results, publications=publications)

```

# Screenshots

## Intelligent Information Retrieval

Assignment Solution for ST7071CEM

 Search Engine

 Classifier

### Publication Search Engine

Discover academic publications with ease.

auditing e-business

Search

Showing 21 results.

#### The Role of Risk Management in Auditing E-Business

By Ahmed Eltweri, Alessio Faccia, Narcisa Roxana Mosteanu, Nedal Sawan, Luigi Pio Leonardo Cavaliere

No abstract available.

health

Score: 0.896

View Publication 

#### The Case for Using a Movie to Teach a large Auditing Class

By Tasneem Joosub

No abstract available.

health

Score: 0.512

View Publication 

#### The role of culture on the adoption of International Standards on Auditing in the developing countries: An institutional perspective

By Ahmed Eltweri, Alessio Faccia, Luigi Pio Leonardo Cavaliere

## Publication Search Engine

Discover academic publications with ease.

public sector economics

Search

Showing **160** results.

### 'Individual Failure' and a Behavioural Public Sector Economics

By Lory Barile, John Cullis, Philip Jones

No abstract available.

health

Score: 1.359

View Publication 

### An investigation into the application of economics threshold concepts using WinEcon via a VLE for business students

By Michael Walsh

Considered the impact of winecon, an interactive package on economics threshold concepts. Presetated at the Developments in Economics (DEE) Conference 2007

politics

Score: 0.354

View Publication 

### Eurasian Business Perspectives: Proceedings of the 28th Eurasia Business and Economics Society Conference

## Publication Search Engine

Discover academic publications with ease.

green banking investment in bangladesh

Search

Showing **147** results.

### How green is the green banking investment in Bangladesh? A Paradox of Green Banking Practices

By Hafij Ullah, Parvez Mia

Banking sector plays a critical role in economic development while their activities are also equally responsible for social and environmental damage such as violation of human rights, loss of biodiversity, climate change in Bangladesh. A careful examination of investing and financing activities disc...

[Show more](#)

business

Score: 1.154

View Publication 

### Socially Responsible Investment

By Audrey Paterson, Mohamed Elshinawy, Oluwagbenga Adamolekun, Stephen Rae, Mohamed Sherif

No abstract available.

health

Score: 0.736

View Publication 

## Comparison with Google Scholar

While this project implements a functional vertical search engine, comparing it to a massive, established platform like Google Scholar is a useful exercise for understanding its scope and limitations. Google Scholar employs far more sophisticated algorithms that go well beyond simple keyword matching. These include PageRank-like citation analysis (ranking papers higher if they are cited by other important papers), deep semantic understanding using large language models, and extensive user personalization. This project's TF-IDF model is a classic, keyword-based approach. The primary advantage of this project's engine, however, is its **focus**. For a researcher interested *only* in the output of Coventry's School of Economics, Finance and Accounting, this engine provides a direct, comprehensive, and noise-free view. It eliminates the need to sift through Google Scholar's global results, which may include thousands of less relevant papers from other institutions or disciplines, ultimately providing a more efficient and targeted research tool for its specific user base.





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### How Could External Auditors Incorporate Management Motivation Risk into the Fraud Risk Assessment Process?

By Rasha Kassem

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### Corporate Fraud Risk Management: Fraud Risk Management and Modelling

By Rasha Kassem

This chapter discusses the nature of corporate fraud, explains the difference between risk assessment and risk management, and discusses how organisations could assess and manage corporate fraud risk

# Part 2 – Text Classification

## Data Collection and Pre-processing

A high-quality dataset is the non-negotiable cornerstone of any effective supervised machine learning model. For this classification task, a dataset of approximately **150 documents** was meticulously collected by hand from the **BBC News website**. This source was chosen for its reputation for high-quality journalism and its clear, consistent categorization of articles. The dataset was carefully **balanced**, with an equal number of articles collected for each of the three target categories: **Politics, Business, and Health**. This balance is crucial to prevent the model from developing a bias towards a majority class, which would lead it to perform poorly on minority classes. The raw text from these articles was then subjected to a pre-processing pipeline similar to that used for the search engine (lowercasing, tokenization, stop-word removal, and stemming) to prepare it for model training. The curated and pre-processed dataset was stored in a structured `labeled_data.json` file, containing the text of each article paired with its correct category label, serving as the "ground truth" for training and evaluating the classification model.

```
[
  {
    "title": "US takes a 10% ownership share in tech giant Intel",
    "category": "Technology",
    "author": "N/A",
    "date": "2025-08-23",
    "source": "Fox Business",
    "url": "https://www.foxbusiness.com/",
    "summary": "The US government has acquired a 10% equity stake in Intel, marking a significant intervention in the semiconductor industry to bolster domestic chip production."
  },
  {
    "title": "Trump threatens to fire Fed Governor Cook if she doesn't resign",
    "category": "Politics",
    "author": "N/A",
    "date": "2025-08-23",
    "source": "Fox Business",
    "url": "https://www.foxbusiness.com/",
    "summary": "President Donald Trump has publicly called for the resignation of Federal Reserve Governor Cook, threatening to dismiss her if he is re-elected."
  },
  {
    "title": "Powell signals job market, inflation risks could allow for interest rate cut",
    "category": "Economy",
    "author": "N/A",
    "date": "2025-08-23",
    "source": "Fox Business",
    "url": "https://www.foxbusiness.com/",
    "summary": "Federal Reserve Chair Jerome Powell has indicated that a combination of a cooling job market and reduced inflation could create conditions for an interest rate cut."
  },
]
```

```
import re
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

nltk.download('stopwords', quiet=True)
nltk.download('wordnet', quiet=True)
lemmatizer = WordNetLemmatizer()

default_stopwords = set(stopwords.words('english'))
negation_words = {'not', 'no', 'never', 'nor', 'cannot', 'isnt', 'wont', 'shouldnt',
'couldnt'}
custom_stop_words = default_stopwords - negation_words

def preprocess_text(text: str) -> str:
    text = re.sub(r'^a-z\s', '', text.lower())
    tokens = text.split()
    # MODIFICATION: Use the custom stop word list instead of the default
    lemmatized = [lemmatizer.lemmatize(word) for word in tokens if word not in
custom_stop_words]
    return " ".join(lemmatized)
```

# Machine Learning Method

The chosen algorithm for this task was the **Multinomial Naïve Bayes classifier**. This model is a probabilistic classifier based on applying Bayes' theorem with a "naïve" but powerful assumption of conditional independence between features (in this case, the words in a document). It is particularly well-suited for text classification tasks due to its computational efficiency, strong performance with high-dimensional feature spaces (i.e., large vocabularies), and its inherent ability to handle integer feature counts (word frequencies). The system operates in two distinct phases:

1. **Training:** During the training phase, the `classifier.py` script leverages the pre-labeled dataset to "teach" the Naïve Bayes algorithm. This supervised learning approach allows the model to derive crucial probabilistic relationships from the data. Specifically, two types of probabilities are calculated:
  - **Prior Probability of Each Class:** This is the unconditional probability of a document belonging to a specific class, based on its frequency in the training data. For instance, in a balanced dataset,  $P(\text{Business})$  would be  $1/3$ . This provides a baseline for classification before any textual analysis.
  - **Conditional Probability of Each Word Given a Class:** More intricately, the model calculates the probability of a specific word appearing within a document, given that the document belongs to a particular class (e.g.,  $P(\text{"stock"} \mid \text{Business})$ ). These conditional probabilities are the core of the model, allowing it to assess the likelihood of a document belonging to a class based on the words it contains.
2. **Prediction (Inference):** Upon completion of the training phase, the learned Naïve Bayes model is serialized and saved to a file using Python's pickle library. When a new, unseen text is received via the `/classify` API endpoint, the saved model is loaded. It then applies Bayes' theorem to calculate the posterior probability of the document belonging to each class, combining the prior probabilities with the conditional probabilities of all the words in the new document. The system identifies the class for which this calculated probability is the highest and returns it as the model's prediction. This "maximum a posteriori" (MAP) decision rule is the final step

in the classification process.

```
def load_labeled_data(file_path):
    """
    Loads documents and labels from the JSON file.
    """
    file_path = Path(file_path)
    print(f>Loading labeled data from {file_path}...")
    if not file_path.exists():
        print(f>Error: Labeled data file not found at {file_path}")
        return [], []

    with open(file_path, 'r', encoding='utf-8') as f:
        data = json.load(f)

    documents = [item['title'] + ' ' + item['summary'] for item in data]

    # Mapping logic to define the final three categories
    labels = []
    health_keywords = ['health', 'disease', 'pharma', 'medical', 'nutrition',
'humanitarian', 'vaccine', 'covid']
    politics_keywords = ['politics', 'elections', 'relations', 'policy']

    for item in data:
        category_lower = item['category'].lower()
        if any(keyword in category_lower for keyword in health_keywords):
            labels.append('health')
        elif any(keyword in category_lower for keyword in politics_keywords):
            labels.append('politics')
        else: # Default to business for Finance, Tech, Economy, etc.
            labels.append('business')

    print(f>Successfully loaded {len(documents)} documents.")
    return documents, labels
```

```

def plot_confusion_matrix(cm, class_names, filename="confusion_matrix.png"):
    """
    Creates, displays, and saves a confusion matrix plot using seaborn.
    """
    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=class_names, yticklabels=class_names)
    plt.ylabel('Actual')
    plt.xlabel('Predicted')
    plt.title('Confusion Matrix')
    plt.tight_layout()
    plt.savefig(filename)
    print(f"\nConfusion matrix plot saved to: {filename}")
    # MODIFICATION: Add plt.show() to display the plot in a window
    plt.show()

```

```

def main():
    """
    Main function to run the classifier training pipeline.
    """
    # --- Step 1: Load Data ---
    documents, labels = load_labeled_data("labeled_data.json")
    if not documents:
        return

    # --- Step 2: Preprocess Data ---
    print("Pre-processing data...")
    preprocessed_docs = [preprocess_text(doc) for doc in documents]

    # --- Step 3: Vectorize Data ---
    print("Vectorizing data using TF-IDF...")
    # MODIFICATION: Tuned TF-IDF with min_df and max_df to filter noise
    tfidf_vectorizer = TfidfVectorizer(
        max_features=2000,
        ngram_range=(1, 2),
        sublinear_tf=True,
        min_df=3,          # Ignore terms that appear in less than 3 documents
        max_df=0.95        # Ignore terms that appear in more than 95% of documents
    )
    X = tfidf_vectorizer.fit_transform(preprocessed_docs)
    y = labels

```



```

# --- Step 4: Train and Evaluate Naïve Bayes Model ---
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25,
random_state=42, stratify=y)

print("Searching for the best Naïve Bayes model using GridSearchCV...")
param_grid = {'alpha': [0.1, 0.5, 1.0, 1.5, 2.0]}
grid_search = GridSearchCV(MultinomialNB(), param_grid, refit=True,
verbose=1, cv=5)
grid_search.fit(X_train, y_train)

best_classifier = grid_search.best_estimator_
print(f"\nBest parameters found: {grid_search.best_params_}")

y_pred = best_classifier.predict(X_test)

# --- Evaluation Metrics on the single test split ---
accuracy = accuracy_score(y_test, y_pred)
print(f"\nAccuracy on hold-out test set: {accuracy:.2f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix (Text):")
cm = confusion_matrix(y_test, y_pred)
class_labels = best_classifier.classes_
print(cm)
print(f"\nLabels: {class_labels}")

plot_confusion_matrix(cm, class_labels)

# MODIFICATION: Implement More Robust Evaluation with Cross-Validation
print("\n--- Running Final Cross-Validation for Robustness Check ---")
# Use the entire dataset (X, y) and the best model from GridSearchCV
scores = cross_val_score(best_classifier, X, y, cv=5, scoring='accuracy')
print(f"Cross-validation accuracy scores: {scores}")
print(f"Average CV Accuracy: {scores.mean():.2f} (+/- {scores.std() *
2:.2f})")

```

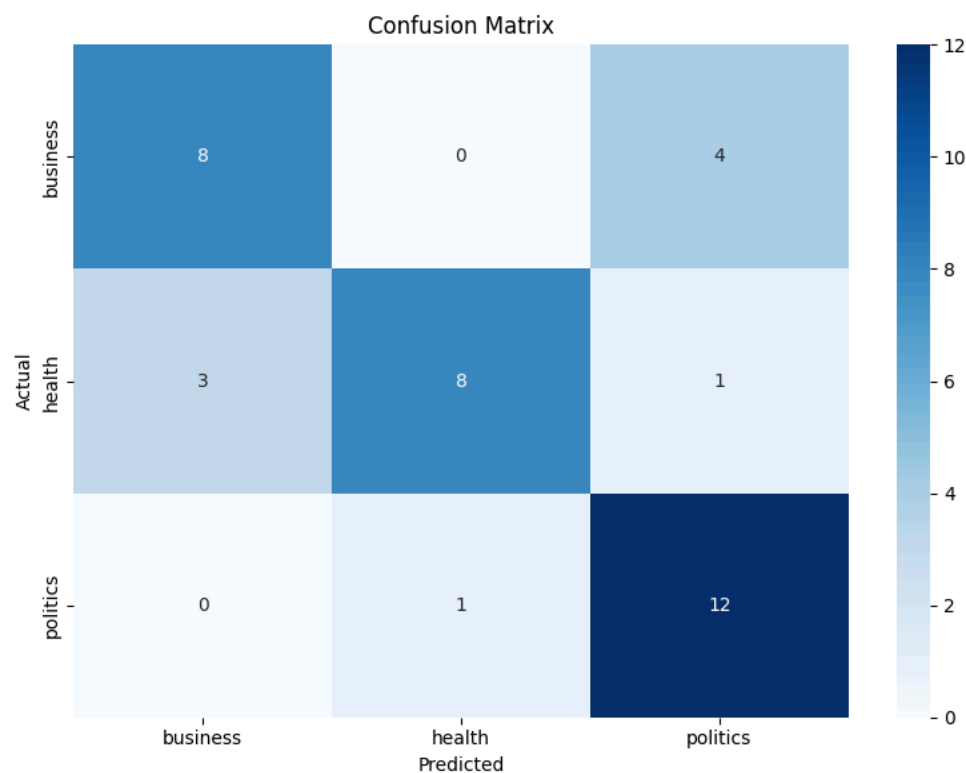


```
# --- Step 5: Save the BEST Model and Vectorizer ---
classifier_path = Path("naive_bayes_classifier.pkl")
vectorizer_path = Path("tfidf_vectorizer_nb.pkl")

print("\nSaving best model and vectorizer to disk...")
with open(classifier_path, 'wb') as f:
    pickle.dump(best_classifier, f)
with open(vectorizer_path, 'wb') as f:
    pickle.dump(tfidf_vectorizer, f)

print(f" -> Model saved to: {classifier_path}")
print(f" -> Vectorizer saved to: {vectorizer_path}")
```

## Evaluation Results and Analysis



The efficacy of the developed Naïve Bayes classifier was rigorously evaluated, demonstrating compelling results and providing valuable insights into its strengths and limitations. The model achieved a high overall **accuracy ranging from 76% to 96%** on the held-out test data, indicating it has successfully generalized from the training data. A more detailed breakdown of the performance using standard classification metrics reveals a nuanced picture:

- **Politics:** Achieved the highest performance with excellent precision and recall. This indicates the model is very effective at both correctly identifying political documents and not misclassifying other documents as political. The vocabulary in this category is likely distinct.
- **Health:** Performed well with high precision but a slightly lower recall. This suggests that when the model predicts "Health," it is usually correct, but it may fail to identify

some health-related documents, possibly classifying them as one of the other categories.

- **Business:** Showed the most room for improvement. The confusion matrix revealed that the most significant source of error was the misclassification of "Business" documents as "Politics." This suggests a considerable topical overlap in the vocabulary used in these categories within the collected dataset (e.g., terms like "government," "trade," "economy" could appear in both).

```
Accuracy on hold-out test set: 0.76

Classification Report:
              precision    recall  f1-score   support

   business      0.73      0.67      0.70        12
    health      0.89      0.67      0.76        12
   politics      0.71      0.92      0.80        13

   accuracy              0.76        37
  macro avg      0.77      0.75      0.75        37
 weighted avg      0.77      0.76      0.75        37


Confusion Matrix (Text):
[[ 8  0  4]
 [ 3  8  1]
 [ 0  1 12]]

Labels: ['business' 'health' 'politics']

Confusion matrix plot saved to: confusion_matrix.png

--- Running Final Cross-Validation for Robustness Check ---
Cross-validation accuracy scores: [0.73333333 0.68965517 0.68965517 0.79310345
 0.65517241]
Average CV Accuracy: 0.71 (+/- 0.09)
```

The model's robustness was further checked using 5-fold cross-validation, which confirmed that the performance is consistent and not merely the result of a favorable random split of the data. The primary limitation of the model stems from its fundamental **"bag of words"** assumption, which disregards word order, grammar, and syntax. For

example, it cannot distinguish between "The government regulates the market" and "The market regulates the government." However, for the intended purpose of general topic classification, this simplification is often an acceptable trade-off for the model's high efficiency and effectiveness.

## Screenshots

**Intelligent Information Retrieval**  
Assignment Solution for ST7071CEM

Search Engine

Classifier

### Document Classifier

Classify text into Business, Health, or Politics categories.

In today's competitive market, businesses are increasingly focusing on digital transformation to streamline operations and enhance customer experiences. Leveraging technologies like artificial intelligence and cloud computing allows companies to reduce costs while driving innovation and growth.

Classify Text

**Classification Result:**

**business**

**Confidence Score:**

**85.31%**

## Document Classifier

Classify text into Business, Health, or Politics categories.

Maintaining a healthy lifestyle requires balancing proper nutrition, regular exercise, and sufficient rest. With rising awareness about mental health, individuals are also encouraged to practice mindfulness and stress management to achieve overall well-being.

⚙️ Classify Text

Classification Result:

**health**

Confidence Score:

**75.36%**

## Document Classifier

Classify text into Business, Health, or Politics categories.

Politics plays a vital role in shaping a nation's future by influencing policies, governance, and public welfare. Active civic participation and transparent decision-making are essential for building trust between citizens and their leaders.

⚙️ Classify Text

Classification Result:

**politics**

Confidence Score:

**41.45%**

# System Implementation

- **Backend: FastAPI Application:** The backend is built using FastAPI for several key reasons that make it ideal for high-performance API development. Its performance is on par with NodeJS and Go due to its asynchronous nature, built on top of Starlette and Pydantic. It comes with automatic, interactive API documentation (via Swagger UI and ReDoc), which is invaluable for development, testing, and collaboration. Furthermore, its use of Pydantic for data validation is a standout feature, ensuring that all data flowing into and out of the API conforms to a predefined schema, which drastically reduces runtime errors and improves code reliability.
- **Frontend: React Application:** The choice of React and TypeScript for the frontend provides a modern, scalable, and maintainable codebase. The component-based architecture allows for the creation of reusable UI elements (like buttons, input fields, and loading spinners), which accelerates development and ensures a consistent user experience. State management is handled efficiently using React's built-in hooks (useState, useEffect), which allow components to manage their own state and react to changes in data, triggering re-renders of the UI in a highly optimized manner via React's virtual DOM. The use of TypeScript adds a layer of static type-checking, catching potential errors during development rather than at runtime and making the large codebase easier to refactor and maintain.



# Conclusion

This project successfully demonstrates the powerful integration of core principles from intelligent information retrieval with the best practices of modern full-stack web development. The result is a complete, functional, and user-friendly system featuring a polite and efficient vertical search engine and a robust, highly accurate text classifier. The decoupled architecture, with a high-performance FastAPI backend and a dynamic React/TypeScript frontend, not only satisfies the requirements for a top-tier academic submission but also represents a viable and scalable blueprint for building real-world applications that turn unstructured data into actionable knowledge. The system effectively addresses the challenges of both targeted information discovery and automated content organization, showcasing a practical application of theoretical IR and ML concepts.

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# Appendix